where $\hat{\boldsymbol{\ell}}_1$ and $\hat{\boldsymbol{\ell}}_2$ are unit vectors along segments *ab* and *cd* and $\mathbf{E}_1$ and $\mathbf{E}_2$ are the electric fields in media 1 and 2. Next, we decompose $\mathbf{E}_1$ and $\mathbf{E}_2$ into components tangential and normal to the boundary (**Fig. 4-20**),

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n}, \tag{4.98a}$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}. \tag{4.98b}$$

Noting that $\hat{\boldsymbol{\ell}}_1 = -\hat{\boldsymbol{\ell}}_2$, it follows that

$$(\mathbf{E}_1 - \mathbf{E}_2) \cdot \hat{\boldsymbol{\ell}}_1 = 0. \tag{4.99}$$

In other words, the component of $\mathbf{E}_1$ along $\hat{\boldsymbol{\ell}}_1$ equals that of $\mathbf{E}_2$ along $\hat{\boldsymbol{\ell}}_1$, for all $\hat{\boldsymbol{\ell}}_1$ tangential to the boundary. Hence,

$$\mathbf{E}_{1t} = \mathbf{E}_{2t} \quad \text{(V/m)}. \tag{4.100}$$

► Thus, the tangential component of the electric field is **continuous** across the boundary between any two media. ◄

Upon decomposing $\mathbf{D}_1$ and $\mathbf{D}_2$ into tangential and normal components (in the manner of Eq. (4.98)) and noting that $\mathbf{D}_{1t} = \varepsilon_1 \mathbf{E}_{1t}$ and $\mathbf{D}_{2t} = \varepsilon_2 \mathbf{E}_{2t}$, the boundary condition on the tangential component of the electric flux density is

$$\frac{\mathbf{D}_{1t}}{\varepsilon_1} = \frac{\mathbf{D}_{2t}}{\varepsilon_2}. \tag{4.101}$$

Next, we apply Gauss's law, as expressed by Eq. (4.29), to determine boundary conditions on the normal components of $\mathbf{E}$ and $\mathbf{D}$. According to Gauss's law, the total outward flux of $\mathbf{D}$ through the three surfaces of the small cylinder shown in **Fig. 4-20** must equal the total charge enclosed in the cylinder. By letting the cylinder's height $\Delta h \to 0$, the contribution to the total flux through the side surface goes to zero. Also, even if each of the two media happens to contain free charge densities, the only charge remaining in the collapsed cylinder is that distributed on the boundary. Thus, $Q = \rho_s \, \Delta s$, and

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_{\text{top}} \mathbf{D}_1 \cdot \hat{\mathbf{n}}_2 \, ds + \int_{\text{bottom}} \mathbf{D}_2 \cdot \hat{\mathbf{n}}_1 \, ds = \rho_s \, \Delta s, \tag{4.102}$$

where $\hat{\mathbf{n}}_1$ and $\hat{\mathbf{n}}_2$ are the outward normal unit vectors of the bottom and top surfaces, respectively. It is important to remember that ***the normal unit vector at the surface of any medium is always defined to be in the outward direction away from that medium.*** Since $\hat{\mathbf{n}}_1 = -\hat{\mathbf{n}}_2$, Eq. (4.102) simplifies to

$$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \quad \text{(C/m}^2\text{)}. \tag{4.103}$$

If D_{1n} and D_{2n} denote as the normal components of $\mathbf{D}_1$ and $\mathbf{D}_2$ along $\hat{\mathbf{n}}_2$, we have

$$D_{1n} - D_{2n} = \rho_s \quad \text{(C/m}^2\text{)}. \tag{4.104}$$

► The normal component of $\mathbf{D}$ changes abruptly at a charged boundary between two different media in an amount equal to the surface charge density. If no charge exists at the boundary, then D_n is continuous across the boundary. ◄

The corresponding boundary condition for $\mathbf{E}$ is

$$\hat{\mathbf{n}}_2 \cdot (\varepsilon_1 \mathbf{E}_1 - \varepsilon_2 \mathbf{E}_2) = \rho_s, \tag{4.105a}$$

or equivalently

$$\varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \rho_s. \tag{4.105b}$$

In summary, (1) the conservative property of $\mathbf{E}$,

$$\nabla \times \mathbf{E} = 0 \quad \leftrightarrow \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \tag{4.106}$$

led to the result that $\mathbf{E}$ has a continuous tangential component across a boundary, and (2) the divergence property of $\mathbf{D}$,

$$\nabla \cdot \mathbf{D} = \rho_v \quad \leftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q, \tag{4.107}$$

led to the result that the normal component of $\mathbf{D}$ changes by ρ_s across the boundary. A summary of the conditions that apply at the boundary between different types of media is given in **Table 4-3**.

其中$\hat{\boldsymbol{\ell}}_1$和$\hat{\boldsymbol{\ell}}_2$是沿线段ab和cd的单位向量，$E_1$和$E_2$是介质1和2中的电场。接下来，我们将$E_1$和$E_2$分解为与边界相切和法向的分量（图4-20），

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n}, \tag{4.98a}$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}. \tag{4.98b}$$

注意到 $\hat{\boldsymbol{\ell}}_1 = -\hat{\boldsymbol{\ell}}_2$，因此

$$(\mathbf{E}_1 - \mathbf{E}_2) \cdot \hat{\boldsymbol{\ell}}_1 = 0. \tag{4.99}$$

换句话说，对于所有与边界相切的 $\hat{\boldsymbol{\ell}}_1$，$E_1$ 沿 $\hat{\boldsymbol{\ell}}_1$ 的分量等于 E_2 沿 $\hat{\boldsymbol{\ell}}_1$ 的分量。因此，

$$\mathbf{E}_{1t} = \mathbf{E}_{2t} \quad \text{(V/m)}. \tag{4.100}$$

► 因此，电场的切向分量在任意两种介质之间的边界上是连续的。 ◄

将 D_1 和 D_2 分解为切向分量和法向分量（按照方程（4.98））并注意到 $D_{1t} = \varepsilon_1 E_{1t}$ 和 $D_{2t} = \varepsilon_2 E_{2t}$，电通量密度切向分量的边界条件为

$$\frac{\mathbf{D}_{1t}}{\varepsilon_1} = \frac{\mathbf{D}_{2t}}{\varepsilon_2}. \tag{4.101}$$

接下来，我们应用高斯定律，如方程所示。（4.29），以确定 E 和 D 法向分量的边界条件。根据高斯定律，D 通过图 4-20 所示小圆柱体三个表面的总向外通量必须等于圆柱体中封闭的总电荷。通过让圆柱体的高度 $\Delta h \to 0$，对通过侧面的总通量的贡献变为零。此外，即使两种介质中的每一种碰巧都含有自由电荷密度，塌缩圆柱体中剩余的唯一电荷也是分布在边界上的电荷。因此，$Q = \rho_s \, \Delta s$，并且

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_{\text{top}} \mathbf{D}_1 \cdot \hat{\mathbf{n}}_2 \, ds + \int_{\text{bottom}} \mathbf{D}_2 \cdot \hat{\mathbf{n}}_1 \, ds = \rho_s \, \Delta s, \tag{4.102}$$

其中 $\hat{n}_1$ 和 $\hat{n}_2$ 分别是底面和顶面的向外法线单位向量。重要的是要记住，任何物体表面的法向单位向量介质始终被定义为远离该介质的向外方向。由于 $\hat{n}_1 = -\hat{n}_2$，等式：（4.102）简化为

$$\hat{\mathbf{n}}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s \quad \text{(C/m}^2\text{)}. \tag{4.103}$$

如果 D_{1n} 和 D_{2n} 表示为 D_1 和 D_2 沿 $\hat{n}_2$ 的法向分量，我们有

$$D_{1n} - D_{2n} = \rho_s \quad \text{(C/m}^2\text{)}. \tag{4.104}$$

► D 的法向分量在两种不同介质之间的带电边界处突然变化，其变化量等于表面电荷密度。如果边界处不存在电荷，则 D_n 在边界上连续。 ◄

E 对应的边界条件为

$$\hat{\mathbf{n}}_2 \cdot (\varepsilon_1 \mathbf{E}_1 - \varepsilon_2 \mathbf{E}_2) = \rho_s, \tag{4.105a}$$

或同等地

$$\varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \rho_s. \tag{4.105b}$$

综上所述，(1) E 的保守性质，

$$\nabla \times \mathbf{E} = 0 \quad \leftrightarrow \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = 0, \tag{4.106}$$

导致 E 具有跨越边界的连续切向分量，以及 (2) D 的发散性质，

$$\nabla \cdot \mathbf{D} = \rho_v \quad \leftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q, \tag{4.107}$$

导致D的法向分量越过边界改变了ρ_s。表 4-3 总结了适用于不同类型介质之间边界的条件。

Table 4-3 Boundary conditions for the electric fields.

Field Component	Any Two Media	Medium 1 Dielectric ε_1	Medium 2 Conductor
Tangential E	$\mathbf{E}_{1t} = \mathbf{E}_{2t}$	$\mathbf{E}_{1t} = \mathbf{E}_{2t} = 0$	
Tangential D	$\mathbf{D}_{1t}/\varepsilon_1 = \mathbf{D}_{2t}/\varepsilon_2$	$\mathbf{D}_{1t} = \mathbf{D}_{2t} = 0$	
Normal E	$\varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \rho_s$	$E_{1n} = \rho_s/\varepsilon_1$	$E_{2n} = 0$
Normal D	$D_{1n} - D_{2n} = \rho_s$	$D_{1n} = \rho_s$	$D_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) normal components of $\mathbf{E}_1$, $\mathbf{D}_1$, $\mathbf{E}_2$, and $\mathbf{D}_2$ are along $\hat{\mathbf{n}}_2$, which is the outward normal unit vector of medium 2.

Example 4-13: Application of Boundary Conditions

The x–y plane is a charge-free boundary separating two dielectric media with permittivities ε_1 and ε_2, as shown in **Fig. 4-21**. If the electric field in medium 1 is

$$E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}}E_{1z},$$

and $\mathbf{E}_1 = \hat{\mathbf{x}}$, find (a) the electric field $\mathbf{E}_2$ in medium 2 and (b) the angles θ_1 and θ_2.

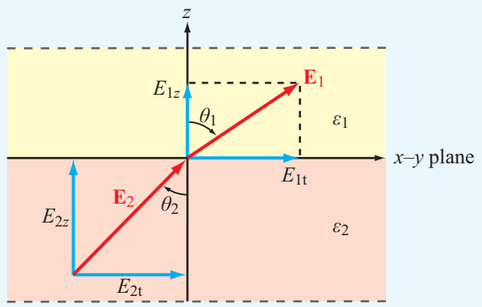


Figure 4-21 Application of boundary conditions at the interface between two dielectric media (Example 4-13).

Solution: (a) Let $\mathbf{E}_2 = \hat{\mathbf{x}}E_{2x} + \hat{\mathbf{y}}E_{2y} + \hat{\mathbf{z}}E_{2z}$. Our task is to find the components of $\mathbf{E}_2$ in terms of the given components of $\mathbf{E}_1$. The normal to the boundary is $\hat{\mathbf{z}}$. Hence, the x and y components of the fields are tangential to the boundary and the z components are normal to the boundary. At a charge-free interface, the tangential components of $\mathbf{E}$ and the normal components of $\mathbf{D}$ are continuous. Consequently,

$$E_{2x} = E_{1x}, \qquad E_{2y} = E_{1y},$$

and

$$D_{2z} = D_{1z} \qquad \text{or} \qquad \varepsilon_2 E_{2z} = \varepsilon_1 E_{1z}.$$

Hence,

$$\mathbf{E}_2 = \hat{\mathbf{x}}E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}}\,\frac{\varepsilon_1}{\varepsilon_2}E_{1z}. \tag{4.108}$$

(b) The tangential components of $\mathbf{E}_1$ and $\mathbf{E}_2$ are

$$E_{1t} = \sqrt{E_{1x}^2 + E_{1y}^2} \qquad \text{and} \qquad E_{2t} = \sqrt{E_{2x}^2 + E_{2y}^2}.$$

The angles θ_1 and θ_2 are then given by

$$\tan\theta_1 = \frac{E_{1t}}{E_{1z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{E_{1z}},$$

$$\tan\theta_2 = \frac{E_{2t}}{E_{2z}} = \frac{\sqrt{E_{2x}^2 + E_{2y}^2}}{E_{2z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{(\varepsilon_1/\varepsilon_2)E_{1z}},$$

and the two angles are related by

$$\frac{\tan\theta_2}{\tan\theta_1} = \frac{\varepsilon_2}{\varepsilon_1}. \tag{4.109}$$

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表 4-3 电场的边界条件。

Field Component	Any Two Media	Medium 1 Dielectric ε_1	Medium 2 Conductor
Tangential E	$\mathbf{E}_{1t} = \mathbf{E}_{2t}$	$\mathbf{E}_{1t} = \mathbf{E}_{2t} = 0$	
Tangential D	$\mathbf{D}_{1t}/\varepsilon_1 = \mathbf{D}_{2t}/\varepsilon_2$	$\mathbf{D}_{1t} = \mathbf{D}_{2t} = 0$	
Normal E	$\varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \rho_s$	$E_{1n} = \rho_s/\varepsilon_1$	$E_{2n} = 0$
Normal D	$D_{1n} - D_{2n} = \rho_s$	$D_{1n} = \rho_s$	$D_{2n} = 0$

Notes: (1) ρ_s is the surface charge density at the boundary; (2) normal components of $\mathbf{E}_1$, $\mathbf{D}_1$, $\mathbf{E}_2$, and $\mathbf{D}_2$ are along $\hat{\mathbf{n}}_2$, which is the outward normal unit vector of medium 2.

例 4-13：边界条件的应用

x-y 平面是一个无电荷边界，分隔了介电常数为 ε_1 和 ε_2 的两种介电介质，如图 4-21 所示。如果介质 1 中的电场为

$$E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}}E_{1z},$$

和 $\mathrm{E}_1 = \hat{\mathrm{x}}$，求 (a) 介质 2 中的电场 E_2 和 (b) 角度 θ_1 和 θ_2 。

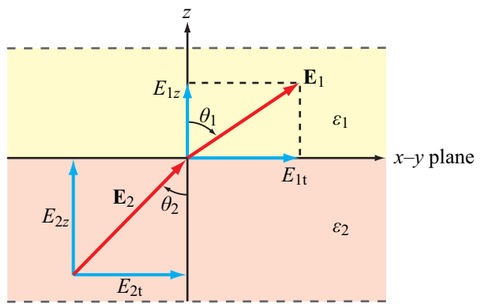


图 4-21 在两种介电介质之间的界面处应用边界条件（示例 4-13）。

解： (a) 令 $\mathrm{E}_2 = \hat{\mathrm{x}}\mathrm{E}_{2x} + \hat{\mathrm{y}}\mathrm{E}_{2y} + \hat{\mathrm{z}}\mathrm{E}_{2z}$。我们的任务是根据给定的分量找到 E_2 的分量

E_1。边界法线为 $\hat{\mathrm{z}}$。因此，场的 x 和 y 分量与边界相切，而 z 分量与边界垂直。在无电荷界面处，E 的切向分量和 D 的法向分量是连续的。最后，

$$E_{2x} = E_{1x}, \qquad E_{2y} = E_{1y},$$

和

$$D_{2z} = D_{1z} \qquad \text{or} \qquad \varepsilon_2 E_{2z} = \varepsilon_1 E_{1z}.$$

因此，

$$\mathbf{E}_2 = \hat{\mathbf{x}}E_{1x} + \hat{\mathbf{y}}E_{1y} + \hat{\mathbf{z}}\,\frac{\varepsilon_1}{\varepsilon_2}E_{1z}. \tag{4.108}$$

(b) E_1 和 E_2 的切向分量为

$$E_{1t} = \sqrt{E_{1x}^2 + E_{1y}^2} \qquad \text{and} \qquad E_{2t} = \sqrt{E_{2x}^2 + E_{2y}^2}.$$

角度 θ_1 和 θ_2 由下式给出

$$\tan\theta_1 = \frac{E_{1t}}{E_{1z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{E_{1z}},$$

$$\tan\theta_2 = \frac{E_{2t}}{E_{2z}} = \frac{\sqrt{E_{2x}^2 + E_{2y}^2}}{E_{2z}} = \frac{\sqrt{E_{1x}^2 + E_{1y}^2}}{(\varepsilon_1/\varepsilon_2)E_{1z}},$$

两个角度的关系为

$$\frac{\tan\theta_2}{\tan\theta_1} = \frac{\varepsilon_2}{\varepsilon_1}. \tag{4.109}$$

Technology Brief 9: Capacitive Sensors

To sense is to respond to a stimulus. (See Technology Brief 7 on resistive sensors.) A capacitor can function as a sensor if the stimulus changes the capacitor's ***geometry***—usually the spacing between its conductive elements—or the effective ***dielectric properties*** of the insulating material situated between them. Capacitive sensors are used in a multitude of applications. A few examples follow.

Fluid Gauge

The two metal electrodes in (**Fig. TF9-1(a)**), usually rods or plates, form a capacitor whose capacitance is directly proportional to the ***permittivity*** of the material between them. If the fluid section is of height h_f and the height of the empty space above it is $(h - h_f)$, then the overall capacitance is equivalent to two capacitors in parallel, or

$$C = C_f + C_a = \varepsilon_f w \frac{h_f}{d} + \varepsilon_a w \frac{(h - h_f)}{d},$$

where w is the electrode plate width, d is the spacing between electrodes, and ε_f and ε_a are the permittivities of the fluid and air, respectively. Rearranging the expression as a linear equation yields

$$C = kh_f + C_0,$$

where the constant coefficient is $k = (\varepsilon_f - \varepsilon_a)w/d$ and $C_0 = \varepsilon_a wh/d$ is the capacitance of the tank when totally empty. Using the linear equation, the fluid height can be determined by measuring C with a bridge circuit (**Fig. TF9-1(b)**).

► The output voltage V_{out} assumes a functional form that depends on the source voltage υ_g, the capacitance C_0 of the empty tank, and the unknown fluid height h_f. ◄

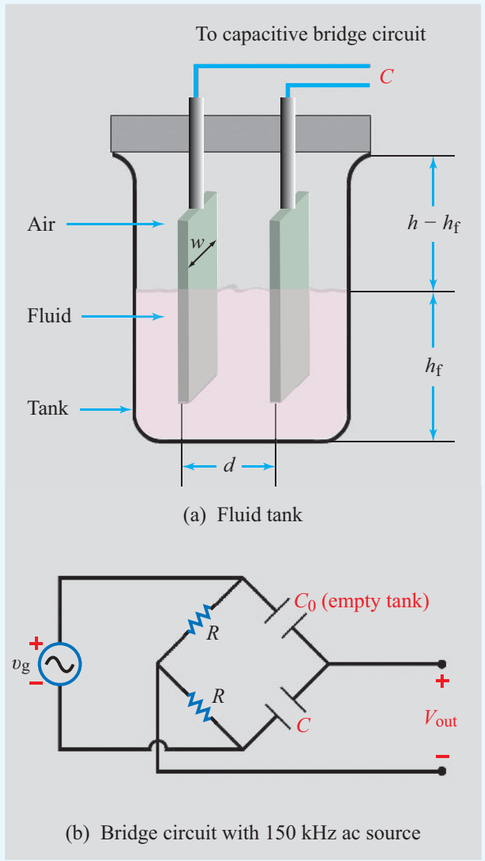


Figure TF9-1 Fluid gauge and associated bridge circuit with C_0 being the capacitance that an empty tank would have and C the capacitance of the tank under test.

Humidity Sensor

Thin-film metal electrodes shaped in an ***interdigitized pattern*** (to enhance the ratio A/d) are fabricated on a silicon substrate (**Fig. TF9-2**). The spacing between digits is typically on the order of $0.2\ \mu m$. *The effective permittivity of the material separating the electrodes varies with the relative humidity of the surrounding environment.* Hence, the capacitor becomes a humidity sensor.

Pressure Sensor

A flexible metal ***diaphragm*** separates an oil-filled chamber with reference pressure P_0 from a second chamber exposed to the gas or fluid whose pressure P is to be measured by the sensor (**Fig. TF9-3(a)**). The membrane is sandwiched—but electrically isolated—between two conductive parallel surfaces, forming two capacitors in series **Fig. TF9-3(b)**. When $P > P_0$, the membrane bends

技术简介 9：电容式传感器

感知就是对刺激做出反应。（请参阅有关电阻传感器的技术简介 7。）如果刺激改变了电容器的几何形状（通常是其导电元件之间的间距）或位于它们之间的绝缘材料的有效介电特性，则电容器可以充当传感器。电容式传感器有多种应用。下面是一些例子。

液位计

（图TF9-1（a））中的两个金属电极（通常是棒或板）形成一个电容器，其电容与它们之间材料的介电常数成正比。如果流体部分的高度为 h_f，其上方空白空间的高度为 $(h - h_f)$，则总电容相当于两个并联电容器，或者

$$C = C_f + C_a = \varepsilon_f w \frac{h_f}{d} + \varepsilon_a w \frac{(h - h_f)}{d},$$

其中 w 是电极板宽度，d 是电极之间的间距，ε_f 和 ε_a 分别是流体和空气的介电常数。将表达式重新排列为线性方程得出

$$C = kh_f + C_0,$$

其中常数系数为 $k = (\varepsilon_f - \varepsilon_a)w/d$，$C_0 = \varepsilon_a wh/d$ 为水箱完全排空时的电容。利用线性方程，可以通过用电桥电路测量C来确定流体高度（图TF9-1（b））。

► 输出电压 V_{out} 呈现出取决于源电压 υ_g、空罐电容 C_0 和未知流体高度 h_f 的函数形式。◄

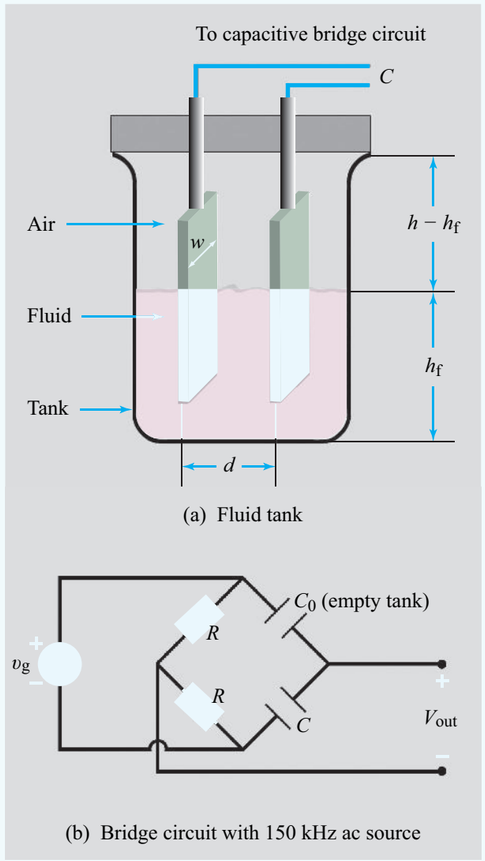


图 TF9-1 液位计和相关电桥电路，其中 C_0 是空罐具有的电容，C 是被测罐的电容。

湿度传感器

在硅基板上制造了呈交叉指状图案形状的薄膜金属电极（以提高比率 A/d）（图 TF9-2）。数字之间的间距通常约为 0.2 μm。分隔电极的材料的有效介电常数随周围环境的相对湿度而变化。因此，电容器成为湿度传感器。

压力传感器

柔性金属隔膜将具有参考压力 P_0 的充油室与暴露于气体或流体的第二个室分开，其压力 P 由传感器测量（图 TF 9-3(a)）。该膜夹在两个导电平行表面之间（但电隔离），形成两个串联电容器（图 TF9-3（b））。当 $P > P_0$ 时，膜弯曲

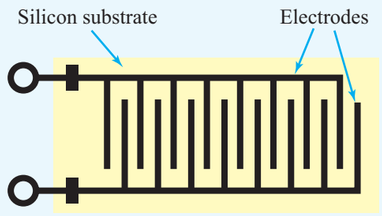


Figure TF9-2 Interdigital capacitor used as a humidity sensor.

in the direction of the lower plate. Consequently, d_1 increases and d_2 decreases, and in turn, C_1 decreases and C_2 increases (**Fig. TF9-3(c)**). The converse happens when $P < P_0$. With the use of a capacitance bridge circuit, such as the one in **Fig. TF9-1(b)**, the sensor can be calibrated to measure the pressure P with good precision.

Noncontact Sensors

Precision positioning is a critical ingredient in semiconductor device fabrication, as well as in the operation and control of many mechanical systems. ***Noncontact capacitive sensors*** are used to sense the position of silicon wafers during the deposition, etching, and cutting processes, without coming in direct contact with the wafers.

▶ Noncontact sensors are also used to sense and control robot arms in equipment manufacturing and to position hard disc drives, photocopier rollers, printing presses, and other similar systems. ◀

The concentric plate capacitor in **Fig. TF9-4** consists of two metal plates sharing the same plane but electrically isolated from each other by an insulating material. When connected to a voltage source, charges of opposite polarity form on the two plates, resulting in the creation of electric field lines between them. The same principle applies to the adjacent plate's capacitor in **Fig. TF9-5**. In both cases, the capacitance is determined by the shapes and sizes of the conductive elements and by the effective permittivity of the dielectric medium containing the electric field lines between them. Often, the capacitor surface is covered by a thin film of nonconductive material, the

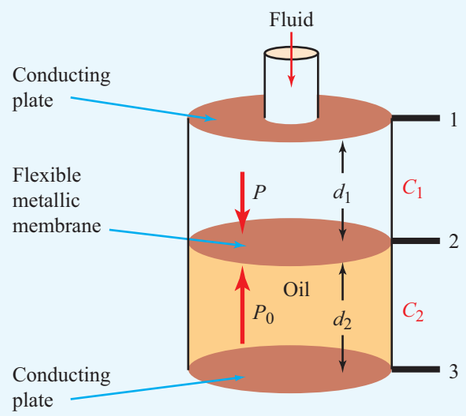
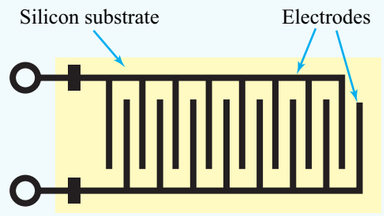


Figure TF9-3 Pressure sensor responds to deflection of metallic membrane.

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图TF9-2 叉指电容用作湿度传感器。

沿下板的方向。因此，d_1 增加，d_2 减少，反过来，C_1 减少，C_2 增加（图 TF9-3(c)）。当 $P < P_0$ 时，情况相反。通过使用电容桥电路，如图TF9-1(b)所示，可以对传感器进行校准，以高精度测量压力P。

非接触式传感器

精确定位是半导体器件制造以及许多机械系统的操作和控制中的关键因素。非接触式电容传感器用于在沉积、蚀刻和切割过程中感测硅晶圆的位置，而无需与晶圆直接接触。

▶ 非接触式传感器还用于感测和控制设备制造中的机器人手臂，以及定位硬盘驱动器、复印机滚筒、印刷机和其他类似系统。 ◀

图TF9-4中的同心板电容器由共享同一平面但通过绝缘材料彼此电隔离的两个金属板组成。当连接到电压源时，两个极板上会形成相反极性的电荷，从而在它们之间产生电场线。同样的原理也适用于图 TF9-5 中相邻板的电容器。在这两种情况下，电容都由导电元件的形状和尺寸以及导电元件之间包含电场线的介电介质的有效介电常数决定。通常，电容器表面覆盖有一层非导电材料薄膜，

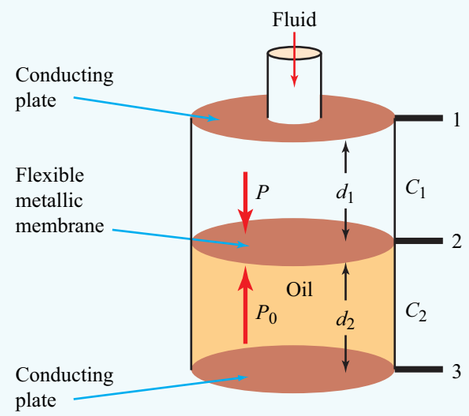


图 TF9-3 压力传感器对金属膜变形的响应。

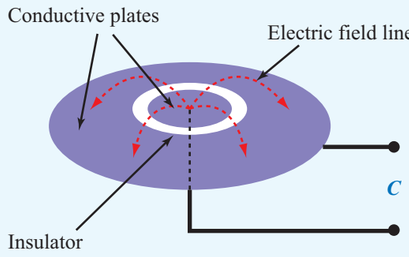
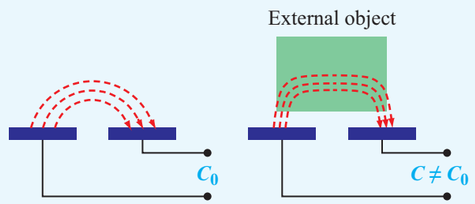


Figure TF9-4 Concentric-plate capacitor.



(a) Adjacent-plates capacitor (b) Perturbation field

Figure TF9-5 (a) Adjacent-plates capacitor; (b) perturbation field.

purpose of which is to keep the plate surfaces clean and dust free.

► The introduction of an external object into the proximity of the capacitor (**Fig. TF9-5(b)**) changes the effective permittivity of the medium, perturbs the electric field lines, and modifies the charge distribution on the plates. ◄

This, in turn, changes the value of the capacitance as it would be measured by a capacitance meter or ***bridge circuit***. Hence, the capacitor becomes a ***proximity sensor***, and its sensitivity depends, in part, on how different the permittivity of the external object is from that of the unperturbed medium and on whether it is or is not made of a conductive material.

Fingerprint Imager

An interesting extension of noncontact capacitive sensors is the development of a fingerprint imager consisting of a two-dimensional array of capacitive sensor cells constructed to record an electrical representation of a fingerprint (**Fig. TF9-6**). Each sensor cell is composed of an adjacent-plates capacitor connected to a capacitance measurement circuit (**Fig. TF9-7**). The entire surface of the imager is covered by a thin layer of nonconductive oxide. When the finger is placed on the oxide surface, it perturbs the field lines of the individual sensor cells to varying degrees, depending on the distance between the ridges and valleys of the finger's surface from the sensor cells.

► Given that the dimensions of an individual sensor are on the order of 65 μm on the side, the imager is capable of recording a fingerprint image at a resolution corresponding to 400 dots per inch or better. ◄

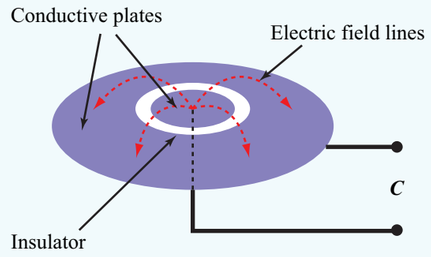
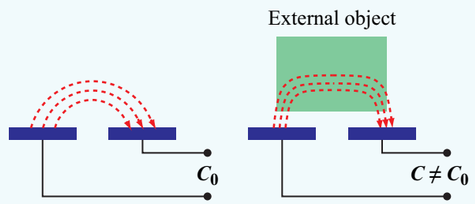


图 TF9-4 同心板电容器。



(a) Adjacent-plates capacitor (b) Perturbation field

图TF9-5(a)邻板电容器； (b) 扰动场。

其目的是保持板材表面清洁无灰尘。

► 将外部物体引入电容器附近（图TF9-5（b））会改变介质的有效介电常数，扰乱电场线，并改变板上的电荷分布。 ◄

这反过来又改变了电容值，因为它是由电容表或电桥电路测量的。因此，电容器成为接近传感器，其灵敏度部分取决于外部物体的介电常数与未受干扰介质的介电常数的差异程度以及它是否由导电材料制成。

指纹成像仪

非接触式电容传感器的一个有趣的扩展是指纹成像仪的开发，该成像仪由二维电容传感器单元阵列组成，用于记录指纹的电气表示（图 TF9-6）。每个传感器单元由连接到电容测量电路的相邻板电容器组成（图 TF9-7）。成像仪的整个表面都覆盖有一层薄薄的非导电氧化物。当手指放在氧化物表面上时，它会不同程度地扰动各个传感器单元的磁力线，具体取决于手指表面的脊和谷与传感器单元之间的距离。

► 鉴于单个传感器的侧面尺寸约为 65 μm，成像器能够以相当于每英寸 400 点或更高的分辨率记录指纹图像。 ◄